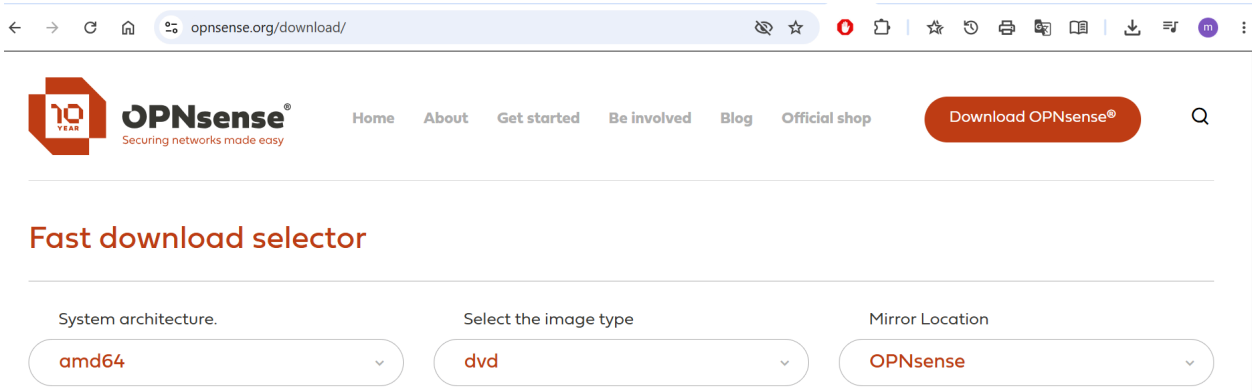
Home Lab Project Week 8, Day 1

Task:

Obtain the pfSense installation image and create the firewall virtual machine in VirtualBox. Allocate virtual hardware and prepare the VM for installation.

We need to go on the website: opnsense.org
Just type OPNsense in your browser and you will see it. Click on it and you are on the homepage.



Let's click on the download button

It takes us to the download page

The version to install is: amd64

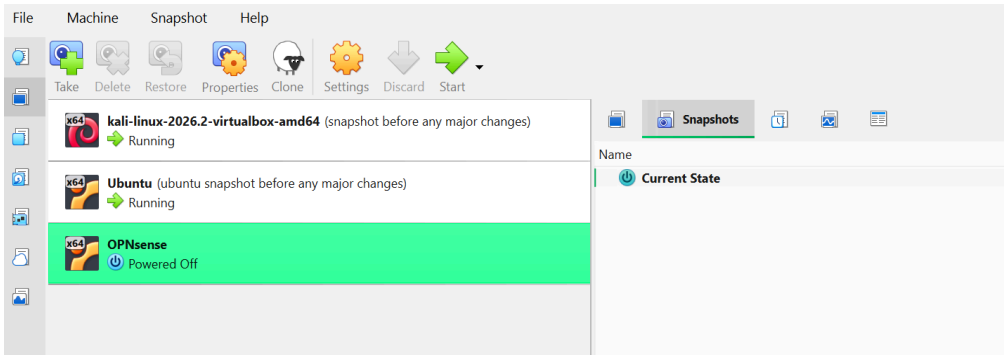
The file will be downloaded as compressed with a .bz2 extension, and we need to extract it.



OPNsense-26.7-dvd-amd64.iso.bz2



OPNsense-26.7-dvd-amd64.iso



VM Creation Settings

Setting	What to choose	Why
OS Type	**BSD**	OPNsense is FreeBSD-based, not Linux
OS Distribution	**FreeBSD**	No OPNsense-specific entry exists; FreeBSD gives correct defaults
OS Version	**FreeBSD (64-bit)**	Matches the 64-bit OPNsense image
Use EFI	☐ **Unchecked**	OPNsense's installer expects traditional BIOS boot, not UEFI
RAM	2 GB+	Comfortable headroom for a firewall VM
CPU	1–2 cores	Lightweight workload

Serial Port	☐ **Unchecked/Disabled**	Only relevant for headless physical appliances; irrelevant in a VM with a normal display

Storage Settings

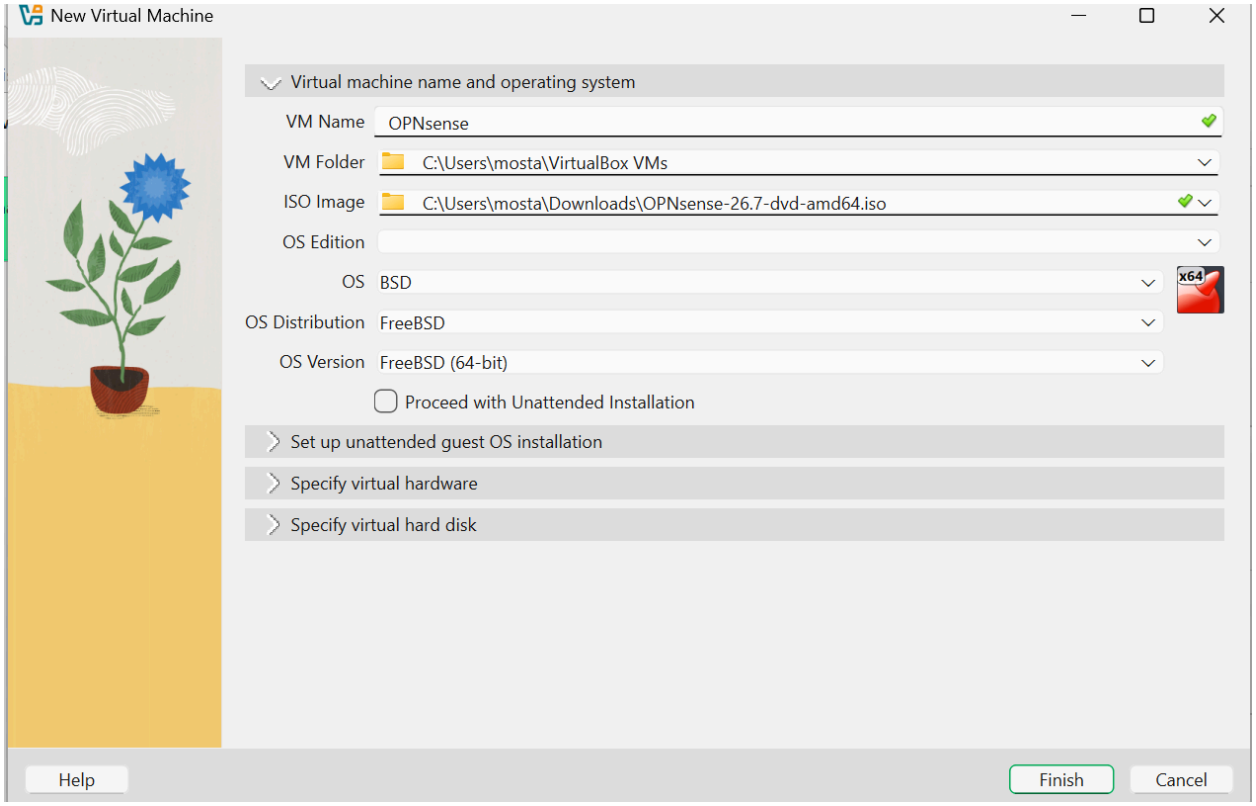
Setting	What to choose	Why
Virtual disk	✅ Created, attached	Where OPNsense actually installs
Installer ISO	✅ Attached to optical drive	Must be the real OPNsense `.iso`, not a `.bz2` wrapper or wrong OS image
Controller type (IDE vs SATA)	Either works	IDE is fine for a lab firewall VM; no real performance concern here

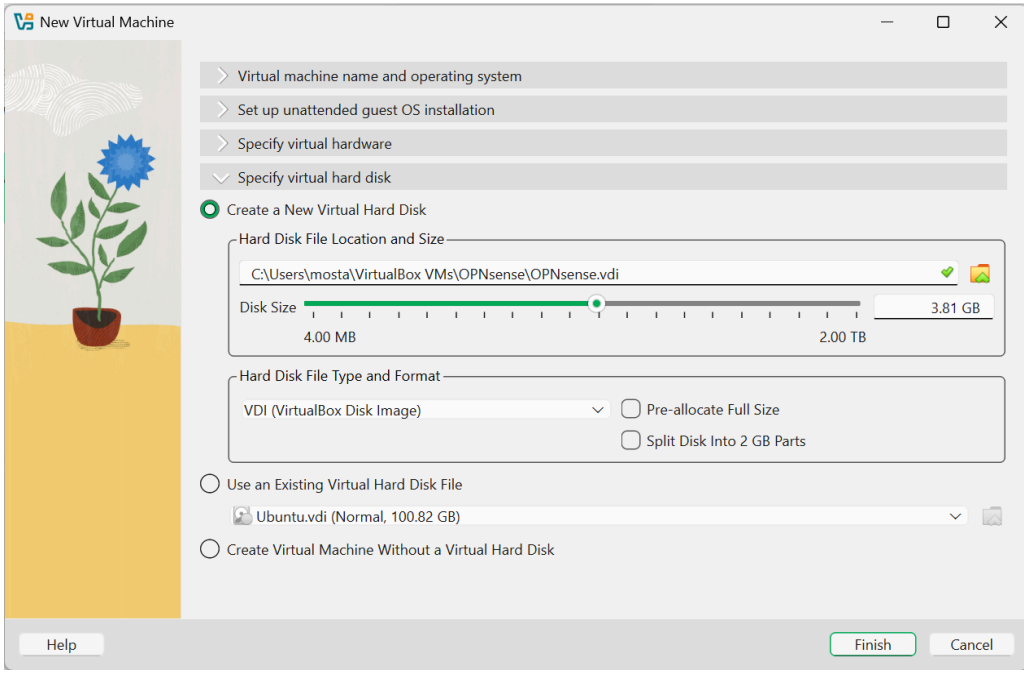
Network Adapter Settings (per VM)

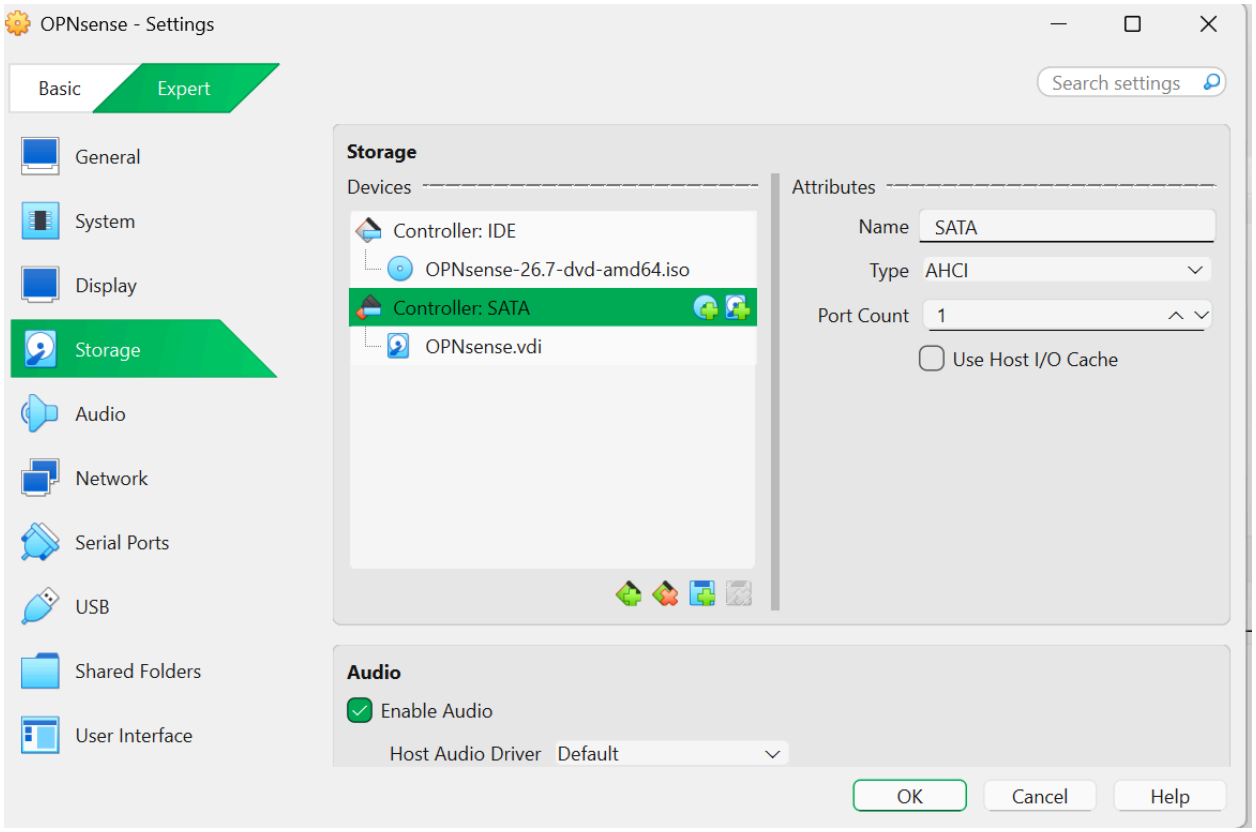
VM	Promiscuous Mode	Why
Kali (attacker, running Wireshark)	✅ **Allow VMs** (or Allow All)	Needed to capture traffic not addressed to Kali itself
OPNsense (firewall)	☐ **Deny** (default)	A firewall sees traffic as the legitimate recipient on each interface; doesn't need promiscuous mode
Ubuntu (target)	☐ **Deny** (default)	Only needs to see its own traffic

Host-Level Windows Settings (fix for VM freezing)

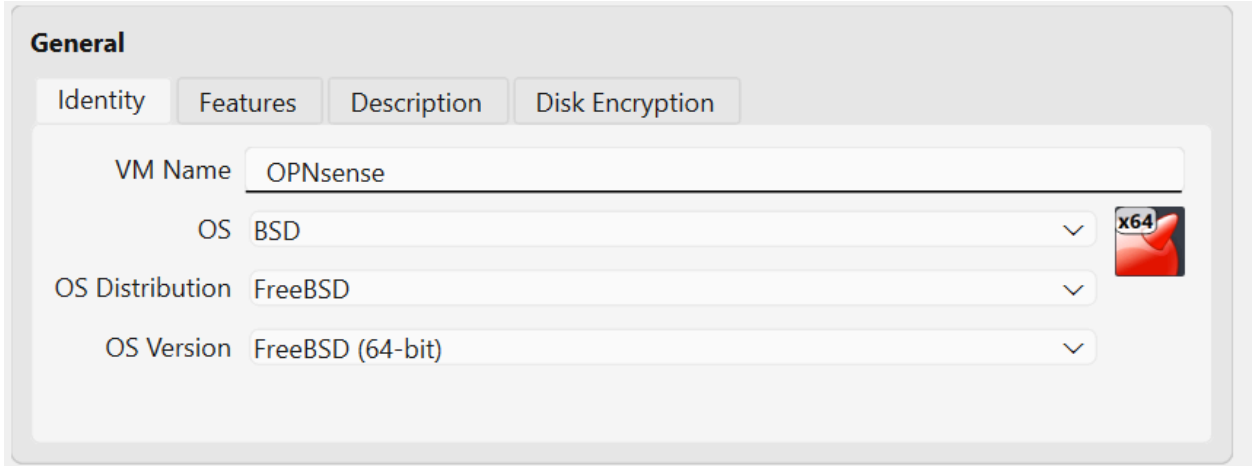
Setting	What to choose	Why
Virtual Machine Platform (Windows Features)	☐ **Unchecked/disabled**	Conflicts with VirtualBox for newer FreeBSD-kernel guests
Windows Hypervisor Platform	☐ **Unchecked/disabled**	Same conflict
Memory Integrity (Core Isolation)	☐ **Off**	Same conflict, affects VM boot reliability

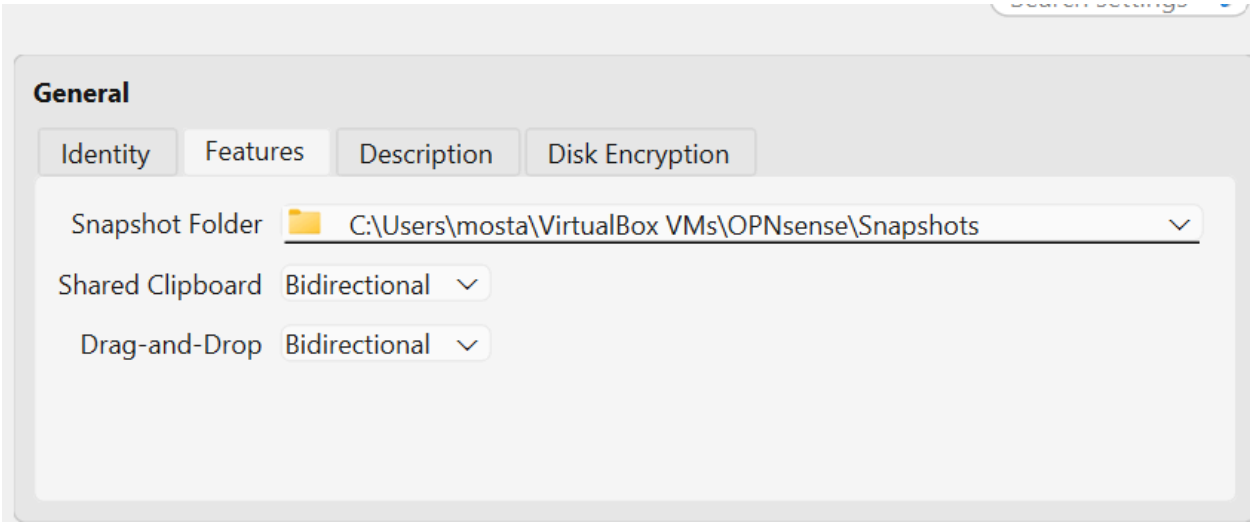

New Virtual Machine
Virtual machine name and operating system
VM Name
OPNsense
VM Folder
C:\Users\mosta\VirtualBox VMs
ISO Image
C:\Users\mosta\Downloads\OPNsense-26.7-dvd-amd64.iso
OS Edition
OS
BSD
OS Distribution
FreeBSD
OS Version
FreeBSD (64-bit)
Proceed with Unattended Installation
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Help
Finish
Cancel

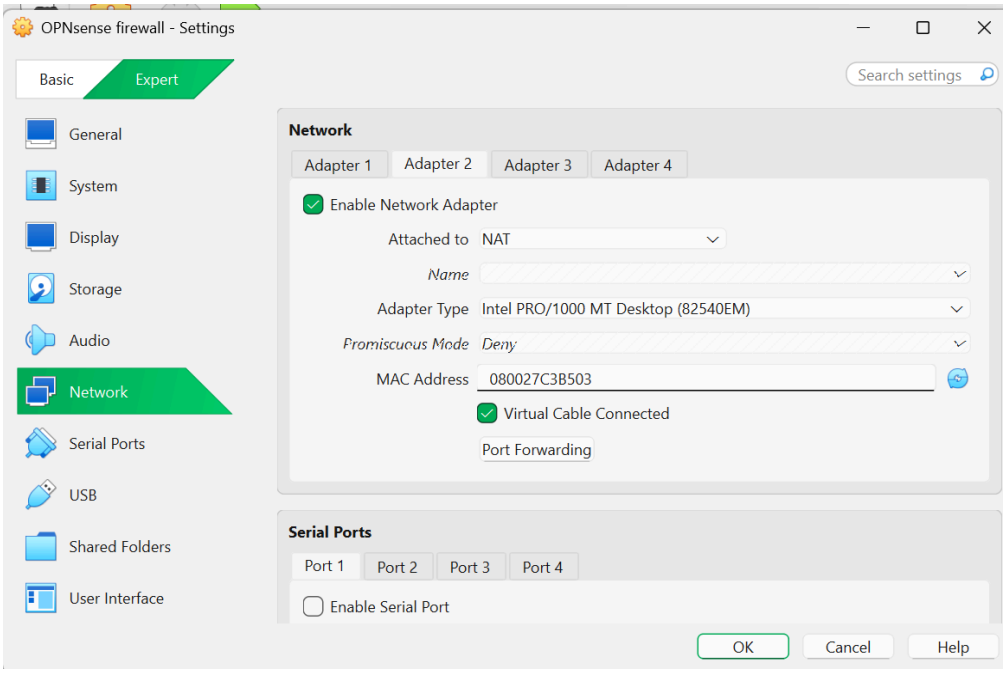

New Virtual Machine
Virtual machine name and operating system
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Create a New Virtual Hard Disk
Hard Disk File Location and Size
C:\Users\mosta\VirtualBox VMs\OPNsense\OPNsense.vdi
Disk Size
3.81 GB
4.00 MB
2.00 TB
Hard Disk File Type and Format
VDI (VirtualBox Disk Image)
Pre-allocate Full Size
Split Disk Into 2 GB Parts
Use an Existing Virtual Hard Disk File
Ubuntu.vdi (Normal, 100.82 GB)
Create Virtual Machine Without a Virtual Hard Disk
Help
Finish
Cancel



- **OS:** Linux → **BSD**
- **OS Distribution:** Ubuntu → look for **FreeBSD** in that dropdown
- **OS Version:** pick **FreeBSD (64-bit)** — there's no OPNsense-specific entry, so generic FreeBSD is correct and gives VirtualBox the right defaults




General
Identity
Features
Description
Disk Encryption
Snapshot Folder
C:\Users\mosta\VirtualBox VMs\OPNsense\Snapshots
Shared Clipboard
Bidirectional
Drag-and-Drop
Bidirectional


OPNsense firewall - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027C3B503
Virtual Cable Connected
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help